

American International University–Bangladesh (AIUB)

Faculty of Science and Technology

**A Unified Forensic Framework for the Detection
of Multimodal Synthetic Media: Identifying
Artifacts in Deepfakes and Generative AI
Content**

Moshin Ibna Hossain (23-50194-1)

Alif Hasan Khan (23-50186-1)

MD. Yeahyea Jam (23-50187-1)

Shadman Shakib (23-50200-1)

A Thesis submitted for the degree of Bachelor of Science (BSc) in
Computer Science and Engineering (CSE) at American International
University–Bangladesh (AIUB), Faculty of Science and Technology (FST)

Summer 2025–2026 Semester

Submission Date: [TODO: Month, Year]

Declaration

This thesis is composed of our original work, and contains no material previously published or written by another person except where due reference has been made in the text. We have clearly stated the contribution of others to our thesis as a whole, including statistical assistance, survey design, data analysis, significant technical procedures, professional editorial advice, financial and any other original research work used or reported in our thesis. The content of our thesis is the result of work we have carried out since the commencement of the Thesis.

We acknowledge that copyright of all material contained in my thesis resides with the copyright holder(s) of that material. Where appropriate we have obtained copyright permission from the copyright holder to reproduce material in this thesis and have sought permission from co-authors for any jointly authored works included in the thesis.

Moshin Ibna Hossain

23-50194-1

Department of Computer Science

Alif Hasan Khan

23-50186-1

Department of Computer Science

MD. Yeahyea Jam

23-50187-1

Department of Computer Science

Shadman Shakib

23-50200-1

Department of Computer Science

Approval

The thesis titled “A Unified Forensic Framework for the Detection of Multimodal Synthetic Media: Identifying Artifacts in Deepfakes and Generative AI Content” has been submitted to the following respected members of the board of examiners of the department of computer science in partial fulfilment of the requirements for the degree of Bachelor of Science in Computer Science on [TODO: date of defense] and has been accepted as satisfactory.

Mohammad Mahmudul Hasan

[TODO: Rank] & Supervisor

Department of Computer Science

American International University–
Bangladesh

[TODO: Name of the External]

[TODO: Rank] & External

Department of Computer Science

American International University–
Bangladesh

Dr. Debajyoti Karmaker

Associate Professor & Head (UG)

Department of Computer Science

American International University–
Bangladesh

Prof. Dr. Dip Nandi

Professor & Associate Dean

Faculty of Science and Technology

American International University–
Bangladesh

Mashiour Rahman

Sr. Associate Professor & Dean

Faculty of Science and Technology

American International University–
Bangladesh

Acknowledgement

The authors express their deepest gratitude to the Almighty for granting the strength, patience, and perseverance required to complete this work.

The authors are sincerely grateful to their supervisor, **Mohammad Mahmudul Hasan**, Department of Computer Science, American International University–Bangladesh, for his continuous guidance, insightful feedback, and encouragement throughout every stage of this thesis. His direction shaped both the research questions and the rigor with which they were pursued.

The authors also thank the Department of Computer Science and the Faculty of Science and Technology of American International University–Bangladesh for providing the academic environment and resources that made this research possible.

No professional editorial service was used in the preparation of this thesis. [TODO: If any editorial advice is received, name the editor and describe the service rendered, as required by the template.]

Author Contributions

The significant and substantial inputs made by the authors to this research, including contributions to the conception and design of the project, non-routine technical work, analysis and interpretation of research data, and drafting or critically revising the work, are listed in Table 1. [TODO: Adjust the contribution descriptions to reflect the team’s actual division of work before submission.]

Table 1: Author contributions.

Full Name	AIUB ID	Contribution
Moshin Ibna Hossain	23-50194-1	Conception and design of the framework; model architecture (DAVID-Net, QACP); training pipeline implementation; drafting of the methodology chapter.
Alif Hasan Khan	23-50186-1	Literature review and gap analysis; dataset acquisition and preprocessing pipeline; drafting of the introduction and existing-studies sections.
MD. Yeahyea Jam	23-50187-1	Experimental evaluation; baseline reproduction; statistical analysis and interpretation of results; drafting of the results chapter.
Shadman Shakib	23-50200-1	Deployment system (inference API and web interface); robustness and impact analysis; critical revision of the full manuscript.

Project-Thesis Planning

The micro tasks related to the management of this thesis are categorized within the work breakdown structure (WBS) shown in Table 2. The available resources comprise the four group members, the supervisor’s weekly consultation hours, an NVIDIA DGX Spark workstation for model training, and free-tier cloud hosting (Hugging Face Spaces, Vercel) for the deployed system. [TODO: Update the Execution Data column as tasks complete; adjust dates to the official semester calendar.]

Table 2: Project-Thesis Deliverables and Logbook.

Task Breakdown	Assigned Person	Schedule Data	Execution Data
Planning and topic finalization	All members	[TODO: date]	[TODO: date]
Literature review and gap analysis	Alif Hasan Khan	[TODO: date]	[TODO: date]
Dataset access requests and collection	Alif Hasan Khan	[TODO: date]	[TODO: date]
Preprocessing pipeline development	Moshin Ibna Hossain	[TODO: date]	[TODO: date]
Baseline model reproduction	MD. Yeahyea Jam	[TODO: date]	[TODO: date]
Proposed model (DAVID-Net + QACP)	Moshin Ibna Hossain	[TODO: date]	[TODO: date]
Experiments and ablation studies	MD. Yeahyea Jam	[TODO: date]	[TODO: date]
Deployment (API and web interface)	Shadman Shakib	[TODO: date]	[TODO: date]
Writing thesis report	All members	[TODO: date]	[TODO: date]
Submission and review	All members	[TODO: date]	[TODO: date]

[TODO: Insert Gantt chart figure (figures/gantt_chart.png) showing scheduled tasks, durations, and start/end times.]

Contents

Declaration	i
Approval	ii
Acknowledgement	iii
Author Contributions	iv
Project-Thesis Planning	v
Contents	vi
List of Figures	viii
List of Tables	ix
List of Abbreviations	x
Abstract	xi
Keywords	xii
1 Introduction	1
1.1 Background Analysis	1
1.2 Existing Studies	2
1.3 Research Motivation and Objective	3
1.4 Research Contribution	4
2 Research Methodology	6
2.1 Conceptual Framework	6
2.1.1 Problem Formulation	6
2.1.2 DAVID-Net Architecture	6
2.1.3 Quadrant-Aware Contrastive Pretraining (QACP)	8

2.1.4	Deployment Architecture	9
2.2	Data Collection	9
2.2.1	Selection of the Data Collection Method	9
2.2.2	Description of the Datasets	9
2.2.3	Preprocessing and Analysis Pipeline	10
2.3	Professional Code of Ethical Issues	10
2.4	Economic Decision	11
3	Results and Analysis	12
3.1	Results and Analysis	12
3.2	Accuracy and Verification	13
3.3	Impact Analysis	13
	Conclusion	15
	Bibliography	16
	Appendix	19
	Funding Agency	20

List of Figures

2.1	Overview of the proposed DAVID-Net architecture. Blue denotes the video branch, orange the audio branch, violet the cross-modal consistency path, and green the multi-task decision heads. The consistency embedding z_c is kept disentangled from the modality-specific authenticity embeddings z_v and z_a	7
2.2	Quadrant-Aware Contrastive Pretraining (QACP). All four authenticity quadrants plus the real-but-mismatched class are constructed from pristine clips alone; three supervised-contrastive losses with different label partitions structure the three embedding spaces (Stage 0), after which the network warm-starts supervised multi-task fine-tuning (Stage 1).	8
2.3	The end-to-end data collection and processing pipeline. Encoder features are cached once (frozen backbones), making every subsequent training run inexpensive and every reported number reproducible from committed manifests and split files.	10

List of Tables

1	Author contributions.	iv
2	Project-Thesis Deliverables and Logbook.	v
1.1	The four authenticity quadrants of a multimodal clip.	1
1.2	Capability comparison of prior work and the proposed framework.	3
2.1	QACP pseudo-classes constructed from pristine clips.	8
2.2	Datasets used in this study.	10
3.1	Per-modality and quadrant results on FakeAVCeleb (subject-disjoint test split). [TODO: fill]	12
3.2	Cross-dataset AUC: train on FakeAVCeleb + AV-Deepfake1M, test unseen. [TODO: fill]	12
3.3	Ablation study. Each row reports in-domain and cross-dataset AUC. [TODO: fill]	13

List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Receiver Operating Characteristic Curve
AV	Audio-Visual
BCE	Binary Cross-Entropy
CNN	Convolutional Neural Network
CS	Computer Science
CSE	Computer Science and Engineering
DSRM	Design Science Research Methodology
ECE	Expected Calibration Error
EER	Equal Error Rate
FVFA	Fake Video, Fake Audio
FVRA	Fake Video, Real Audio
GAN	Generative Adversarial Network
IoU	Intersection over Union
LOGO	Leave-One-Generator-Out
QACP	Quadrant-Aware Contrastive Pretraining
RVFA	Real Video, Fake Audio
RVRA	Real Video, Real Audio
SSL	Self-Supervised Learning
SupCon	Supervised Contrastive Learning
TTS	Text-to-Speech
VC	Voice Conversion
WBS	Work Breakdown Structure

Abstract

The purpose of this research is to address a critical limitation of current synthetic-media forensics: existing audio-visual deepfake detectors reduce a rich, multimodal problem to a single binary “real versus fake” decision, discarding forensically essential information about *which* modality of a video was manipulated and *when*. Because modern generative artificial intelligence can synthesize facial video and cloned speech independently, a clip may contain a genuine face with a cloned voice, a swapped face over genuine speech, or a fully synthetic combination of both; distinguishing these cases is necessary for journalism, content moderation, and legal contexts.

This thesis proposes a unified forensic framework that reformulates the task as fine-grained, per-modality attribution. Given a video clip, the framework independently determines whether the visual stream and the acoustic stream are authentic or AI-generated, assigns the clip to one of four authenticity quadrants, and temporally localizes the manipulated segments in each stream. At its core is DAVID-Net, a disentangled dual-branch cross-modal transformer that separates modality-specific authenticity evidence from cross-modal consistency evidence, enabling the detection of internally consistent double-fakes that defeat synchronization-based detectors. To generalize to unseen generators, the study introduces Quadrant-Aware Contrastive Pretraining (QACP): a self-supervised stage that constructs all four authenticity quadrants from pristine data alone—using self-blended face video, vocoder copy-synthesis audio, and cross-clip mismatch—and structures the embedding space with a factorized supervised-contrastive objective, without using any deepfake generator outputs. The framework is evaluated on FakeAVCeleb, AV-Deepfake1M, and LAV-DF, with cross-dataset validation on DFDC and KoDF, using per-modality AUC, four-class attribution accuracy, temporal localization average precision, calibration, robustness, and fairness metrics.

[TODO: Summarize the primary findings in two to three sentences once experiments are complete.]

The findings would be helpful for forensic analysts, fact-checking organizations, and online platforms that must not only flag manipulated media but also explain which component was fabricated. The complete system is released as reproducible open-source software and deployed as a public web service comprising a FastAPI inference API and a Next.js user interface, allowing end users to verify whether the video and the voice of a clip are genuine or AI-generated.

Keywords

deepfake detection, multimodal forensics, audio-visual learning, synthetic media, contrastive pretraining, temporal localization, generative ai, cross-modal consistency

CHAPTER 1

INTRODUCTION

1.1 Background Analysis

The rapid advancement of generative artificial intelligence has transformed the creation of synthetic media from a specialized research activity into a commodity capability. Face-swapping, facial reenactment, lip-synchronization models, text-to-speech (TTS), and voice conversion (VC) systems are publicly available, and their outputs are increasingly indistinguishable from genuine recordings by human observers. International bodies and platform operators have repeatedly warned that manipulated audio-visual content is now used for disinformation, financial fraud, and non-consensual impersonation, making reliable detection a societal necessity rather than a purely academic exercise.

The root cause of the detection problem is that the visual and acoustic streams of a video can be manipulated *independently*. A single clip can therefore fall into one of four authenticity quadrants, summarized in Table 1.1: both streams real (RVRA), a cloned voice dubbed over real footage (RVFA), a swapped or reenacted face over genuine speech (FVRA), or a fully synthetic combination (FVFA).

Table 1.1: The four authenticity quadrants of a multimodal clip.

Quadrant	Video	Audio	Typical origin
RVRA	Real	Real	Genuine recording
RVFA	Real	Fake	Voice clone / TTS dubbed over real footage
FVRA	Fake	Real	Face swap / reenactment over real speech
FVFA	Fake	Fake	Fully synthetic avatar (lip-sync + TTS)

This problem is important because the overwhelming majority of published detectors collapse it into a single binary decision. Such a formulation is forensically inadequate for three reasons. First, it provides no *attribution*: a journalist, court, or moderation system needs to know whether the face was swapped, the voice was cloned, or both. Second, binary detectors tend to shortcut on whichever modality is easier to classify, degrading accuracy on the harder stream. Third, modern manipulations are increasingly *partial*—only a few seconds of audio or a short facial segment are altered—which a clip-level binary label cannot express. Solving per-modality attribution with temporal localization therefore places this thesis in the correct context of the wider synthetic-media forensics problem domain.

1.2 Existing Studies

Five research threads are directly relevant to this study.

Unimodal video detection. Early detectors treated the task as frame-level image classification: XceptionNet trained on FaceForensics++ [1] became the canonical baseline, followed by EfficientNet ensembles that won the Deepfake Detection Challenge [2]. Temporal methods such as LipForensics [3] and RealForensics [4] model mouth-motion dynamics for stronger cross-manipulation generalization. These methods are blind to audio manipulation and degrade sharply under compression and dataset shift.

Unimodal audio spoofing detection. The anti-spoofing community, organized around the ASVspoof challenges [5], progressed from spectral features to end-to-end models such as RawNet2 [6] and the spectro-temporal graph attention network AASIST [7]; self-supervised speech front-ends such as WavLM [8] further improved robustness. Audio-only methods cannot detect facial manipulation.

Audio-visual binary detection. Multimodal detectors fuse both streams but almost universally emit one label: the Modality Dissonance Score [9], affective-cue matching [10], and transformer fusion models such as AVFakeNet [11] and AVoiD-DF [12]. A real-video/fake-audio clip and a fake-video/real-audio clip receive the same output.

Consistency and localization. SyncNet [13]-style synchronization was exploited for anomaly-based detection [14], but consistency methods detect *mismatch* without attributing which stream is fake and fail on fully synthetic clips whose fake streams are mutually consistent. LAV-DF [15] and AV-Deepfake1M [16] introduced temporal forgery localization, but localization is usually joint over modalities and cross-dataset generalization is under-reported.

Proxy-forgery training. Self-Blended Images [17] and Face X-ray [18] showed that synthesizing blending artifacts from pristine images generalizes better than training on generator outputs; the anti-spoofing literature found the same for vocoder copy-synthesis [19]. Zhou and Lim [20] explored joint audio-visual labels but trained on generator outputs without a pretraining stage.

Research gap. As summarized in Table 1.2, no existing method simultaneously provides (a) independent per-modality authenticity, (b) cross-modal consistency reasoning, (c) temporal localization, and (d) validated cross-generator generalization in a single reproducible, deployable system. This study utilizes the strongest components of the existing literature (self-supervised encoders, contrastive synchronization, proxy forgeries) and extends them to close exactly this gap.

Table 1.2: Capability comparison of prior work and the proposed framework.

Approach	Per-modality	Consistency	Localization	Cross-gen. eval
Unimodal video	–	–	partial	partial
Unimodal audio	–	–	partial	partial
AV binary fusion	–	partial	–	partial
Sync/anomaly based	–	✓	partial	partial
Localization (BA-TFD)	partial	partial	✓	–
Proposed framework	✓	✓	✓	✓

1.3 Research Motivation and Objective

The general aim of this thesis is to contribute a forensic framework that moves synthetic-media detection from coarse binary classification to fine-grained, per-modality attribution with temporal localization, and to demonstrate that such attribution can generalize to generators never seen during training.

The specific objectives are:

- to design a unified framework that detects, per modality, whether the video and the audio of a clip are real or AI-generated;
- to propose DAVID-Net, a disentangled cross-modal transformer with independent per-modality decision heads, a synchronization-consistency module, and per-modality temporal localization;
- to propose Quadrant-Aware Contrastive Pretraining (QACP), a generator-free pre-training strategy for cross-generator generalization;
- to evaluate the framework across multiple benchmarks with cross-dataset, robustness, calibration, and fairness protocols; and
- to deploy the detector as a publicly accessible web application.

The following research questions (RQs) are answered through the findings of this research:

- RQ1.** Can a single model produce independent, well-calibrated authenticity decisions for the visual and acoustic streams while remaining competitive with specialized unimodal detectors?
- RQ2.** Does explicit modeling of audio-visual synchronization improve detection of manipulations that break cross-modal coherence?

- RQ3.** Does disentangling modality-specific authenticity evidence from cross-modal consistency evidence improve generalization to unseen generators and datasets?
- RQ4.** Can the model temporally localize which segments of each stream were manipulated, and does this objective aid the clip-level decision?
- RQ5.** Is the detector robust to real-world degradations and fair across demographic subgroups?
- RQ6.** Can an embedding space pretrained on synthetically constructed quadrants built from pristine data alone generalize to real forgeries from unseen generators better than supervised training on generator outputs?

1.4 Research Contribution

The target users of the proposed solution are forensic analysts, journalists and fact-checking organizations, online platform moderation systems, and ordinary end users who need to verify received media. They benefit from verdicts that state *which* stream was manipulated and *when*, with calibrated confidence, rather than an unexplained binary flag; the deployed web interface makes this capability directly accessible without technical expertise.

The contribution of this thesis to the development of scientific concepts is:

- **QACP:** a self-supervised pretraining stage that constructs all four authenticity quadrants from pristine data only and structures three embedding spaces with a factorized supervised-contrastive objective, including an explicit *real-but-mismatched* class that teaches the model that desynchronization is not forgery;
- a reformulation of audio-visual deepfake detection as a multi-task, per-modality attribution and localization problem;
- **DAVID-Net**, a disentangled dual-branch cross-modal transformer architecture implementing this reformulation;
- a generalization-first evaluation protocol with leave-one-generator-out and cross-dataset testing as first-class results; and
- a complete, reproducible, publicly deployed detection system.

The remainder of this thesis is organized as follows. Chapter 2 presents the research methodology, including the conceptual framework (DAVID-Net and QACP), the data collection process, ethical considerations, and the economic analysis. Chapter 3 reports the

results, their verification, and an impact analysis. The final chapter concludes the thesis and outlines directions for future work.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter justifies and explains the methodological approach of the study. The research follows the Design Science Research Methodology (DSRM): the problem is identified from the literature (Chapter 1), an artifact (the detection framework) is designed and developed, and the artifact is demonstrated and evaluated against explicit research questions using established benchmarks. DSRM is appropriate because the contribution of this thesis is a designed computational artifact whose utility must be demonstrated empirically, rather than a purely observational study.

2.1 Conceptual Framework

The life cycle of the project comprises five phases: (1) data acquisition and preprocessing, (2) generator-free pretraining (QACP), (3) supervised multi-task training of DAVID-Net, (4) evaluation against baselines and ablations, and (5) deployment as a public web service. The proposed framework is trained in two stages: **Stage 0**, Quadrant-Aware Contrastive Pretraining on synthetically constructed quadrants; and **Stage 1**, supervised multi-task fine-tuning of the full DAVID-Net architecture.

2.1.1 Problem Formulation

Let a clip $x = (x_v, x_a)$ consist of a visual stream x_v and an acoustic stream x_a . The framework predicts:

$$\hat{y}_v = P(\text{video fake} \mid x) \in [0, 1], \quad (2.1)$$

$$\hat{y}_a = P(\text{audio fake} \mid x) \in [0, 1], \quad (2.2)$$

$$\hat{q} \in \{\text{RVRA}, \text{RVFA}, \text{FVRA}, \text{FVFA}\}, \quad (2.3)$$

$$\hat{m}_v(t), \hat{m}_a(t) \in [0, 1] \quad (\text{per-frame manipulation probability}). \quad (2.4)$$

2.1.2 DAVID-Net Architecture

The overall architecture is shown in Figure 2.1. DAVID-Net is built on the observation that “is stream X internally authentic?” and “do the two streams agree with each other?” are different questions. Fusing them prematurely allows shortcut learning (e.g., labeling every

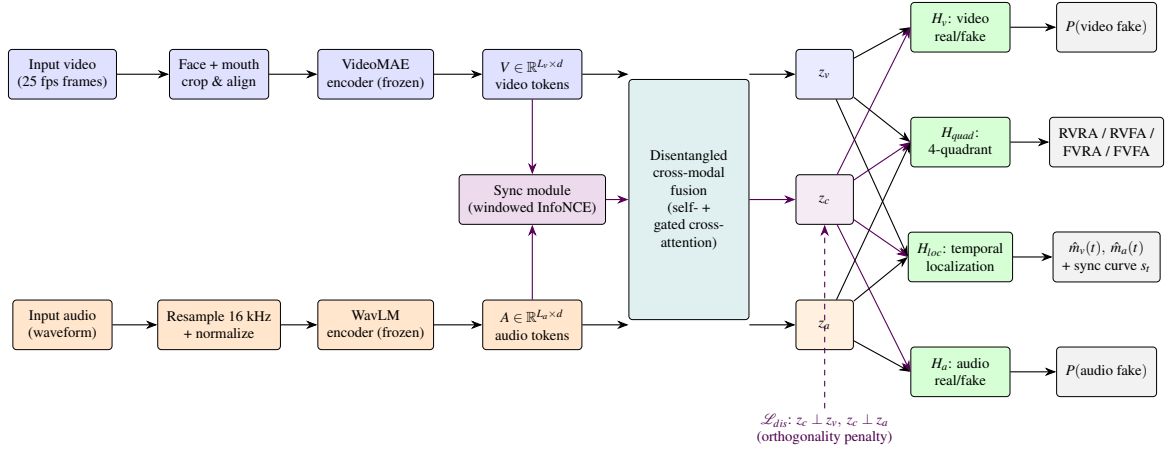


Figure 2.1: Overview of the proposed DAVID-Net architecture. Blue denotes the video branch, orange the audio branch, violet the cross-modal consistency path, and green the multi-task decision heads. The consistency embedding z_c is kept disentangled from the modality-specific authenticity embeddings z_v and z_a .

desynchronized clip fake). The architecture therefore keeps *modality-specific authenticity* evidence disentangled from *cross-modal consistency* evidence, combining them only at the decision heads. This design lets the per-modality heads fire independently—making all four quadrants reachable, including FVFA clips whose two fake streams are mutually consistent.

Encoders. The video branch extracts aligned face and mouth crops and encodes them with a self-supervised VideoMAE backbone into a spatio-temporal token sequence $V \in \mathbb{R}^{L_v \times d}$. The audio branch encodes the raw 16 kHz waveform with WavLM [8] into $A \in \mathbb{R}^{L_a \times d}$. Encoders are largely frozen; only adapters and the fusion layers are trained heavily.

Synchronization-consistency module. Short overlapping windows of V and A are projected into a shared space and trained with an InfoNCE contrastive objective in which temporally aligned real windows are positives. The module outputs a per-window agreement score s_t and a pooled consistency embedding z_c .

Disentangled cross-modal fusion. A cross-modal transformer produces pooled authenticity embeddings z_v, z_a and the consistency embedding z_c . A disentanglement penalty discourages leakage between the authenticity and consistency subspaces:

$$\mathcal{L}_{\text{dis}} = |\cos(z_v, z_c)| + |\cos(z_a, z_c)|. \quad (2.5)$$

Multi-task heads and objective. Four heads consume the disentangled representations: video authenticity $H_v([z_v; z_c])$, audio authenticity $H_a([z_a; z_c])$, quadrant $H_{\text{quad}}([z_v; z_a; z_c])$, and per-frame localization H_{loc} . The total Stage-1 objective is

$$\mathcal{L} = \lambda_1 \mathcal{L}_v + \lambda_2 \mathcal{L}_a + \lambda_3 \mathcal{L}_{\text{quad}} + \lambda_4 \mathcal{L}_{\text{loc}} + \lambda_5 \mathcal{L}_{\text{sync}} + \lambda_6 \mathcal{L}_{\text{dis}}, \quad (2.6)$$

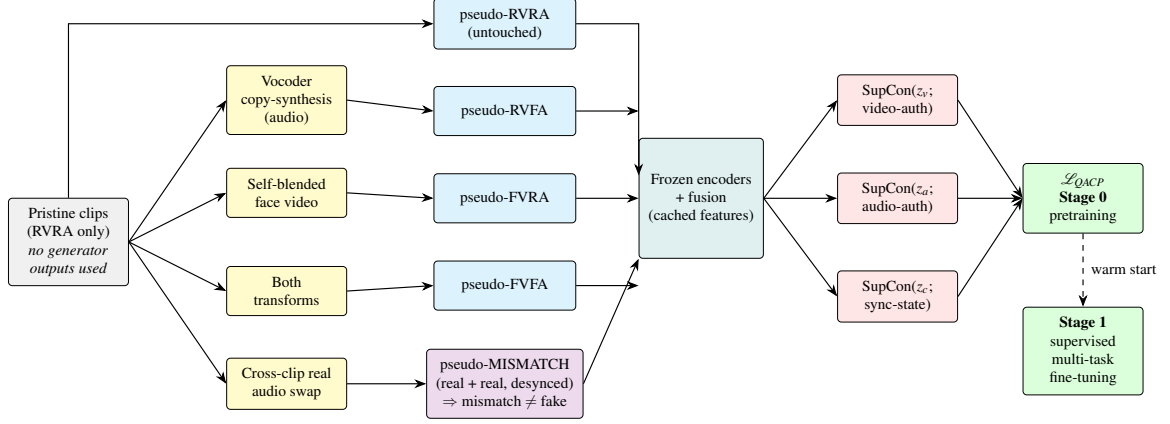


Figure 2.2: Quadrant-Aware Contrastive Pretraining (QACP). All four authenticity quadrants plus the real-but-mismatched class are constructed from pristine clips alone; three supervised-contrastive losses with different label partitions structure the three embedding spaces (Stage 0), after which the network warm-starts supervised multi-task fine-tuning (Stage 1).

where $\mathcal{L}_v, \mathcal{L}_a$ are focal binary cross-entropy losses, $\mathcal{L}_{quad}$ is cross-entropy, $\mathcal{L}_{loc}$ is frame-level binary cross-entropy against manipulated-interval masks, and $\mathcal{L}_{sync}$ is the contrastive synchronization loss applied to genuine (RVRA) samples.

2.1.3 Quadrant-Aware Contrastive Pretraining (QACP)

Detectors trained on generator outputs inherit those generators’ fingerprints and fail on unseen synthesis methods. QACP instead pretrains the embedding space on forgeries *constructed from pristine data alone*, so the space encodes what manipulation does to a stream rather than what a particular generator looks like. Starting from verified-real (RVRA) clips only, five pseudo-classes are manufactured (Table 2.1); the complete two-stage training flow is illustrated in Figure 2.2.

Table 2.1: QACP pseudo-classes constructed from pristine clips.

Pseudo-class	Construction	Teaches
pseudo-RVRA	untouched real clip	anchor
pseudo-RVFA	vocoder copy-synthesis of the audio	audio authenticity cues
pseudo-FVRA	self-blended face video [17]	video authenticity cues
pseudo-FVFA	both transforms	consistent double-fake
pseudo-MISMATCH	real audio swapped from another clip	mismatch $\neq$ fake

The MISMATCH class is the disentanglement forcing function: both streams remain genuinely real while synchronization is broken, so the authenticity heads must answer “real” while the consistency embedding flags disagreement. Three supervised-contrastive (SupCon [21]) losses are applied to the three embedding spaces, each with a *different* label

partition of the same batch:

$$\mathcal{L}_{\text{QACP}} = \mathcal{L}_{\text{SupCon}}(z_v; \text{video-auth}) + \mathcal{L}_{\text{SupCon}}(z_a; \text{audio-auth}) + \mathcal{L}_{\text{SupCon}}(z_c; \text{sync-state}). \quad (2.7)$$

Because each space is supervised to be invariant to the other factors, the factorization itself performs disentanglement.

2.1.4 Deployment Architecture

The complete framework is deployed as a two-tier system: a FastAPI inference service hosting a lightweight DAVID-Net variant on Hugging Face Spaces, and a Next.js web interface that presents per-modality verdicts, quadrant attribution, temporal manipulation timelines, and the synchronization curve to end users.

2.2 Data Collection

2.2.1 Selection of the Data Collection Method

This research uses *established benchmark datasets* rather than an interview, survey, or systematic literature review as its primary data source. This choice follows directly from the research questions: RQ1–RQ6 concern the measurable behavior of a computational artifact on labeled audio-visual media, which only curated, ground-truth-annotated corpora can answer. Interviews or surveys measure human opinion and cannot supply the frame-level manipulation labels required for training and for the localization task; benchmark corpora additionally allow direct, reproducible comparison against prior work under identical splits, which is the accepted evaluation standard in this field.

2.2.2 Description of the Datasets

Table 2.2 lists the datasets used. FakeAVCeleb [22] provides native four-quadrant labels and is the primary training set. AV-Deepfake1M [16] and LAV-DF [15] drive the temporal-localization task. DFDC [2] and KoDF are reserved exclusively for cross-dataset evaluation. ASVspoof [5] strengthens the audio branch. All datasets are obtained under their respective end-user license agreements.

Table 2.2: Datasets used in this study.

Dataset	Modality	Labels	Role
FakeAVCeleb	A+V	4-quadrant	train + in-domain test
AV-Deepfake1M	A+V	clip + localization	train + localization
LAV-DF	A+V	localized forgeries	localization
DFDC	A+V	binary	cross-dataset test only
KoDF	A+V	binary	cross-dataset test only
ASVspoof 2019/21	A	bona fide / spoof	audio-branch auxiliary

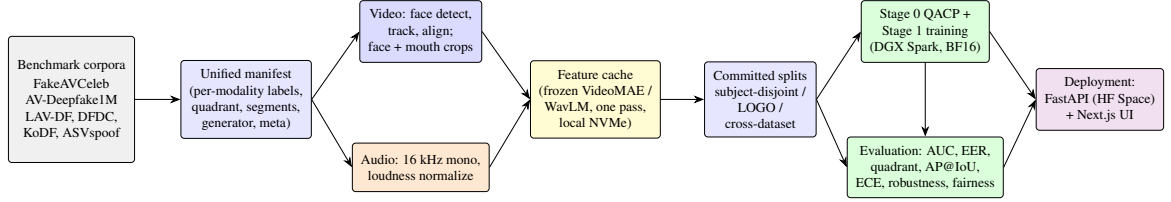


Figure 2.3: The end-to-end data collection and processing pipeline. Encoder features are cached once (frozen backbones), making every subsequent training run inexpensive and every reported number reproducible from committed manifests and split files.

2.2.3 Preprocessing and Analysis Pipeline

Video is decoded at 25 fps; faces are detected, tracked, and aligned, producing a 224×224 face crop and a 96×96 mouth crop. Audio is resampled to 16 kHz mono and loudness-normalized. All datasets are converted to a unified manifest schema with per-modality labels, quadrant, manipulated intervals, generator identity, and demographic metadata, making the analysis reproducible from committed configuration files. Splits are subject-disjoint; leave-one-generator-out (LOGO) splits hold out one manipulation method entirely. The end-to-end data pipeline, from raw corpora to the deployed service, is shown in Figure 2.3.

Experiments are implemented in PyTorch and executed on an NVIDIA DGX Spark system (GB10 Grace-Blackwell, 128 GB unified memory) using BF16 precision and a cached-feature training regime. Evaluation uses per-modality AUC, EER, macro-F1, four-class quadrant accuracy, localization AP@IoU, expected calibration error, robustness curves under compression and noise, and fairness gaps across demographic subgroups; statistical rigor is ensured through three-seed runs, bootstrap confidence intervals, and DeLong tests. Baselines include Xception [1], LipForensics [3], RawNet2 [6], AASIST [7], MDS [9], and AVoiD-DF [12], evaluated under identical splits.

2.3 Professional Code of Ethical Issues

The ethical dimension of this research parses into four sub-problems, each with a concrete mitigation.

Consent and licensing of biometric data. The datasets contain faces and voices of real individuals. All corpora are used strictly under their end-user license agreements, no raw media is redistributed, and only manifests, split definitions, and model weights are released.

Sampling bias and fairness. Benchmark corpora over-represent certain demographics and languages, which can bias detection accuracy. The study audits performance gaps across gender, apparent skin tone, and language subgroups, applies balanced sampling during training, and reports the residual gaps transparently rather than concealing them.

Transparency and research integrity. All reported numbers correspond to committed configuration files with fixed random seeds; negative and partial results of ablations are reported alongside positive ones, guarding against falsification, fabrication, and selective reporting. There is no conflict of interest and no external funding influence on the findings.

Dual-use and responsible deployment. The system is strictly defensive: no generation capability is included or released. The public interface presents verdicts as calibrated probabilities with an explicit “probabilistic, not proof” disclaimer, user uploads are not retained beyond the request, and the model card documents intended use and limitations.

2.4 Economic Decision

Two alternative strategies were compared for the experimental infrastructure. **Alternative A (cloud GPU rental):** training on rented A100-class instances at current market rates, with an estimated total of [TODO: X] GPU-hours across all runs, ablations, and seeds. **Alternative B (owned DGX Spark workstation):** a one-time hardware cost with negligible marginal cost per experiment. Because the experimental plan requires 15–20 full training runs plus continuous development iteration, the owned-hardware alternative amortizes favorably within a single semester and removes the risk of budget-driven experiment truncation; it was therefore selected. The cached-feature training regime further reduces compute cost by amortizing the expensive encoder forward passes into a one-time preprocessing step.

For real-world deployment, the cost breakdown of the proposed solution is deliberately minimal: the demonstration model is served from a free-tier Hugging Face Space (CPU) with an optional GPU upgrade ([TODO: quote current rate]); the web interface is hosted on Vercel’s free tier; and the heavyweight model can be self-hosted on the same DGX Spark workstation at an estimated [TODO: X] W average power draw. A platform-scale adopter would instead face inference costs of approximately [TODO: estimate] per thousand clips on commodity cloud GPUs, which the lightweight distilled variant is specifically designed to reduce.

CHAPTER 3

RESULTS AND ANALYSIS

3.1 Results and Analysis

[TODO: This section is populated as experiments complete. The table skeletons below define the empirical evidence for each research question.]

The proposed solution answers the research questions through four families of evidence: in-domain attribution (RQ1), cross-dataset and cross-generator generalization (RQ3, RQ6), temporal localization (RQ4), and module ablations (RQ2, RQ6). The basic functionality of the proposed solution—independent per-modality verdicts with quadrant attribution and manipulation timelines—makes use of state-of-the-art self-supervised encoders and the QACP pretraining strategy introduced in Chapter 2.

Table 3.1: Per-modality and quadrant results on FakeAVCeleb (subject-disjoint test split).
[TODO: fill]

Method	Video AUC	Audio AUC	Quadrant Acc.	Macro-F1	EER
Xception (video-only)	—	—	—	—	—
AASIST (audio-only)	—	—	—	—	—
MDS	—	—	—	—	—
AVoiD-DF	—	—	—	—	—
Proposed (ours)	—	—	—	—	—

Table 3.2: Cross-dataset AUC: train on FakeAVCeleb + AV-Deepfake1M, test unseen.
[TODO: fill]

Method	DFDC	KoDF	DF-TIMIT
Best unimodal baseline	—	—	—
Best multimodal baseline	—	—	—
Proposed w/o QACP	—	—	—
Proposed + QACP	—	—	—

[TODO: Add temporal localization table (AP@IoU on AV-Deepfake1M and LAV-DF vs. BA-TFD+), qualitative figures (attention heatmaps, synchronization curves, localization timelines), and a discussion connecting each result to its RQ.]

Table 3.3: Ablation study. Each row reports in-domain and cross-dataset AUC. [TODO: fill]

Configuration	In-domain AUC	Cross-dataset AUC
Full framework (+ QACP)	—	—
— QACP pretraining	—	—
— MISMATCH pseudo-class	—	—
— synchronization module	—	—
— disentanglement loss	—	—
early fusion	—	—
late fusion	—	—
single-task (binary only)	—	—
— localization head	—	—

3.2 Accuracy and Verification

[TODO: Fill after experiments.] The accuracy of the findings is verified through four mechanisms, each establishing a direct connection between the collected research data and the research objective. First, *statistical verification*: every headline metric is reported as the mean and standard deviation over three seeds with bootstrap 95% confidence intervals, and AUC differences against baselines are tested with DeLong tests under Holm–Bonferroni correction. Second, *calibration*: expected calibration error and reliability diagrams verify that reported confidences are trustworthy, which is essential for forensic use. Third, *robustness*: detection quality is measured under H.264 re-compression, audio noise, and resolution loss to identify potential points of failure, and the observed degradation profile defines the operating envelope of the system. Fourth, *external validity*: cross-dataset and leave-one-generator-out protocols verify that the solution is resilient to distribution shift rather than fitted to one benchmark. Trade-offs between the full research model and the lightweight deployed variant (accuracy versus latency and hosting cost) are presented to reflect realistic product requirements.

3.3 Impact Analysis

Societal impact. Reliable per-modality attribution strengthens the information ecosystem: fact-checkers can state precisely which component of a viral clip was fabricated, platforms can apply proportionate moderation, and citizens gain a free verification tool. Conversely, detector outputs could be misused to dismiss authentic evidence (“liar’s dividend”); the system mitigates this by reporting calibrated probabilities with explicit uncertainty rather than categorical verdicts.

Legal and safety issues. Detection outputs are decision support, not legal proof; the

interface and model card state this explicitly. The framework’s localization output supports human expert review, which remains the appropriate standard in legal contexts.

Cultural and fairness issues. Because training corpora over-represent some demographics and languages, the fairness audit (Section 3.2) quantifies subgroup gaps, and balanced sampling reduces them; residual gaps are documented so downstream users understand for whom the detector is less reliable.

Sustainability. The cached-feature training regime and the lightweight deployed variant minimize energy consumption relative to end-to-end training and serving of large models. The QACP construction is self-refreshing—new pseudo-forgeries can be synthesized from pristine data as generators evolve—so the solution remains maintainable without repeated large-scale data collection, benefiting stakeholders from platform operators to end users.

Conclusion

This thesis addressed the inadequacy of binary deepfake detection in a media landscape where the visual and acoustic streams of a clip can be manipulated independently. The stated objective—to detect, per modality, whether the video and the voice of a clip are real or AI-generated, to attribute the clip to one of four authenticity quadrants, and to localize manipulated segments in time—was pursued through a designed artifact combining DAVID-Net, a disentangled cross-modal transformer, with Quadrant-Aware Contrastive Pretraining, a generator-free pretraining strategy that constructs all four quadrants from pristine data alone.

The data collection and analysis process relied on established benchmark corpora (FakeAVCeleb, AV-Deepfake1M, LAV-DF) with cross-dataset validation on unseen corpora (DFDC, KoDF), unified under a single reproducible manifest schema with subject-disjoint and leave-one-generator-out splits. [TODO: Summarize the principal findings and their significance in one paragraph once experiments are complete.] The complete system is released as open-source software and deployed as a public web service.

Limitations. The framework targets single-subject talking-head content; multi-person scenes, overlapping speech, and non-speech audio are out of scope. Detection outputs are probabilistic decision support, not proof of authenticity. [TODO: Add empirical limitations discovered during experimentation.]

Future work. Future researchers can extend this study toward fully generated text-to-video content without a real source subject; continual adaptation to emerging generators using the QACP construction as a self-refreshing training signal; multi-person and non-frontal scenarios; integration with provenance standards such as C2PA content credentials; and on-device, real-time variants of the detector.

Bibliography

- [1] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, “Face-forensics++: Learning to detect manipulated facial images,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.
- [2] B. Dolhansky, J. Bitton, B. Pflaum, J. Lu, R. Howes, M. Wang, and C. Canton Ferrer, “The deepfake detection challenge (DFDC) dataset,” *arXiv preprint arXiv:2006.07397*, 2020.
- [3] A. Haliassos, K. Vougioukas, S. Petridis, and M. Pantic, “Lips don’t lie: A generalisable and robust approach to face forgery detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
- [4] A. Haliassos, R. Mira, S. Petridis, and M. Pantic, “Leveraging real talking faces via self-supervision for robust forgery detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [5] X. Wang, J. Yamagishi, M. Todisco, H. Delgado, A. Nautsch, N. Evans, M. Sahidullah, V. Vestman, T. Kinnunen, K. A. Lee, *et al.*, “ASVspoof 2019: A large-scale public database of synthesized, converted and replayed speech,” *Computer Speech & Language*, vol. 64, p. 101114, 2020.
- [6] H. Tak, J. Patino, M. Todisco, A. Nautsch, N. Evans, and A. Larcher, “End-to-end anti-spoofing with RawNet2,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021.
- [7] J.-w. Jung, H.-S. Heo, H. Tak, H.-j. Shim, J. S. Chung, B.-J. Lee, H.-J. Yu, and N. Evans, “AASIST: Audio anti-spoofing using integrated spectro-temporal graph attention networks,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022.
- [8] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, Z. Chen, J. Li, N. Kanda, T. Yoshioka, X. Xiao, *et al.*, “WavLM: Large-scale self-supervised pre-training for full stack speech processing,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, pp. 1505–1518, 2022.
- [9] K. Chugh, P. Gupta, A. Dhall, and R. Subramanian, “Not made for each other: Audio-visual dissonance-based deepfake detection and localization,” in *Proceedings of the ACM International Conference on Multimedia*, 2020.

- [10] T. Mittal, U. Bhattacharya, R. Chandra, A. Bera, and D. Manocha, “Emotions don’t lie: An audio-visual deepfake detection method using affective cues,” in *Proceedings of the ACM International Conference on Multimedia*, 2020.
- [11] H. Ilyas, A. Javed, and K. M. Malik, “AVFakeNet: A unified end-to-end dense swin transformer deep learning model for audio-visual deepfakes detection,” *Applied Soft Computing*, vol. 136, p. 110124, 2023.
- [12] W. Yang, X. Zhou, Z. Chen, B. Guo, Z. Ba, Z. Xia, X. Cao, and K. Ren, “AVoid-DF: Audio-visual joint learning for detecting deepfake,” *IEEE Transactions on Information Forensics and Security*, vol. 18, pp. 2015–2029, 2023.
- [13] J. S. Chung and A. Zisserman, “Out of time: Automated lip sync in the wild,” in *ACCV Workshops*, 2016.
- [14] C. Feng, Z. Chen, and A. Owens, “Self-supervised video forensics by audio-visual anomaly detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.
- [15] Z. Cai, K. Stefanov, A. Dhall, and M. Hayat, “Do you really mean that? content driven audio-visual deepfake dataset and multimodal method for temporal forgery localization,” in *International Conference on Digital Image Computing: Techniques and Applications (DICTA)*, 2022.
- [16] Z. Cai, S. Ghosh, A. P. Adatia, M. Hayat, A. Dhall, and K. Stefanov, “AV-Deepfake1M: A large-scale LLM-driven audio-visual deepfake dataset,” in *Proceedings of the ACM International Conference on Multimedia*, 2024.
- [17] K. Shiohara and T. Yamasaki, “Detecting deepfakes with self-blended images,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [18] L. Li, J. Bao, T. Zhang, H. Yang, D. Chen, F. Wen, and B. Guo, “Face X-ray for more general face forgery detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [19] X. Wang and J. Yamagishi, “Spoofed training data for speech spoofing countermeasure can be efficiently created using neural vocoders,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.
- [20] Y. Zhou and S.-N. Lim, “Joint audio-visual deepfake detection,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

- [21] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, and D. Krishnan, “Supervised contrastive learning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [22] H. Khalid, S. Tariq, M. Kim, and S. S. Woo, “Fakeavceleb: A novel audio-video multimodal deepfake dataset,” *NeurIPS Datasets and Benchmarks Track*, 2021.

Appendix

A. Repository and Reproducibility

The complete source code, configuration files, and dataset split manifests are available at: <https://github.com/MIHMahmudEli/david-net-av>. Every reported number corresponds to a committed configuration file; random seeds are fixed and environments pinned.

B. Unified Manifest Schema

Every clip in every dataset is converted to the following record:

```
{
  "clip_id": "str",
  "video_label": 0,           // 0 real, 1 fake
  "audio_label": 1,
  "quadrant": "RVFA",
  "video_segments": [[s, e]], // manipulated intervals (sec)
  "audio_segments": [[s, e]],
  "generator": "wav2lip",
  "dataset": "fakeavceleb",
  "meta": {"gender": "...", "language": "..."}
}
```

C. API Response Format

The deployed detection service returns:

```
{
  "video": {"verdict": "fake", "confidence": 0.94},
  "audio": {"verdict": "real", "confidence": 0.88},
  "quadrant": {"label": "FVRA", "probs": {...}},
  "localization": {"video": [{"start": 1.2, "end": 3.6,
                                "score": 0.91}], "audio": []},
  "sync_curve": [...],
  "model_version": "david-net-lite-1.0"
}
```

Funding Agency

The authors have received no funding for this research.